

DEPARTMENT OF MATHEMATICS, IIT Guwahati
MA102: Mathematics II
Tutorial Problems Sheet

37. Examine whether the following maps $T : V \rightarrow W$ are linear transformations.

- (a) $V = W = \mathbb{C}^2, \mathbb{F} = \mathbb{C}$ and $T[z_1, z_2]^t = [\bar{z}_1, \bar{z}_2]^t$ for all $[z_1, z_2]^t \in \mathbb{C}^2$.
(b) $V = W = M_n(\mathbb{R}), \mathbb{F} = \mathbb{R}$ and fix $B \in M_n(\mathbb{R})$. Consider $T(A) = AB - BA$ for all $A \in V$.

Solution:

- (a) T is not a linear transformation, as

$$\begin{bmatrix} -i \\ 0 \end{bmatrix} = T\left(\begin{bmatrix} i \\ 0 \end{bmatrix}\right) = T\left(i \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) \neq iT\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} i \\ 0 \end{bmatrix}.$$

- (b) Let $X, Y \in M_n(\mathbb{R})$ and $a, b \in \mathbb{R}$. We have

$$\begin{aligned} T(aX + bY) &= (aX + bY)B - B(aX + bY) \\ &= a(XB - BX) + b(YB - BY) \\ &= aT(X) + bT(Y). \end{aligned}$$

Hence T is a linear transformation.

□

38. Let $T : V \rightarrow V$ be a linear transformation such that $T \circ T = I$ and let $\mathbf{v} \in V$.

- (a) Show that $\{\mathbf{v}, T(\mathbf{v})\}$ is linearly dependent if and only if $T(\mathbf{v}) = \pm\mathbf{v}$.
(b) Give an example of such a linear transformation with $V = \mathbb{R}^2$.

Solution:

- (a) If $T(\mathbf{v}) = \pm\mathbf{v}$, then it is clear that $\{\mathbf{v}, T(\mathbf{v})\}$ is linearly dependent.

Conversely, suppose that $\{\mathbf{v}, T(\mathbf{v})\}$ is linearly dependent. If $\mathbf{v} = \mathbf{0}$ then clearly $T(\mathbf{v}) = \pm\mathbf{v}$. If $\mathbf{v} \neq \mathbf{0}$, then $T(\mathbf{v}) = a\mathbf{v}$ for some $a \in \mathbb{F}$. We have

$$\begin{aligned} T(\mathbf{v}) = a\mathbf{v} &\Rightarrow T(T(\mathbf{v})) = T(a\mathbf{v}) \Rightarrow T^2(\mathbf{v}) = aT(\mathbf{v}) \\ &\Rightarrow \mathbf{v} = a^2\mathbf{v} \\ &\Rightarrow (1 - a^2)\mathbf{v} = \mathbf{0} \\ &\Rightarrow 1 - a^2 = 0, \quad \text{since } \mathbf{v} \neq \mathbf{0} \\ &\Rightarrow a = \pm 1 \\ &\Rightarrow T(\mathbf{v}) = \pm\mathbf{v}. \end{aligned}$$

- (b) Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T([x, y]^t) = [-x, -y]^t$ for all $[x, y]^t \in \mathbb{R}^2$.

□

39. Find bases for the range space and the null space of the linear transformation $T : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ defined by $T(A) = AB - BA$ for all $A \in M_2(\mathbb{R})$, where $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Hence find the rank and the nullity of T .

Solution: Let $A = \begin{bmatrix} x & y \\ z & w \end{bmatrix} \in \ker(T)$ (the null space of T). Then we have

$$\begin{aligned} T(A) = \mathbf{0} &\Rightarrow AB - BA = \mathbf{0} \\ &\Rightarrow \begin{bmatrix} x & y \\ z & w \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} z - y & w - x \\ x - w & y - z \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ &\Rightarrow z - y = 0, w - x = 0, x - w = 0, y - z = 0 \\ &\Rightarrow y = z, x = w. \end{aligned}$$

Thus $A = \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & y \\ y & x \end{bmatrix} = x \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + y \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Also, the set $\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ is linearly independent. Hence $\mathcal{B}$ is a basis for $\ker(T)$.

Now let $X \in \text{range}(T)$. Then $X = T(A) = AB - BA$, for some $A = \begin{bmatrix} x & y \\ z & w \end{bmatrix} \in M_2(\mathbb{R})$. We have

$$X = AB - BA = \begin{bmatrix} z - y & w - x \\ x - w & y - z \end{bmatrix} = (z - y) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + (w - x) \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Also, the set $\mathcal{C} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$ is linearly independent. Hence $\mathcal{C}$ is a basis for $\text{range}(T)$.

We see that $\text{nullity}(T) = 2$ and $\text{rank}(T) = 2$.

Alternative: Consider the standard basis $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ of $M_2(\mathbb{R})$.

Then $\{T(E_{11}), T(E_{12}), T(E_{21}), T(E_{22})\}$ will span $\text{range}(T)$. Now reduce this spanning set to a basis of $\text{range}(T)$. $\square$

40. Let V and W be two finite-dimensional vector spaces and let $T : V \rightarrow W$ be a linear transformation. Show that

- (a) if $\dim(V) < \dim(W)$ then T is not onto; and
- (b) if $\dim(V) > \dim(W)$ then T is not one-one.

Solution: Let $\dim(V) = n$ and $\dim(W) = m$.

- (a) Let T be onto. Then we have $\text{range}(T) = W$, and so $\text{rank}(T) = \dim(W) = m$. We have

$$\text{rank}(T) + \text{nullity}(T) = n \Rightarrow \text{nullity}(T) = n - m \geq 0 \Rightarrow n \geq m.$$

- (b) Let T be one-one. Then we have $\ker(T) = \{\mathbf{0}\}$, and so $\text{nullity}(T) = 0$. We have

$$\text{rank}(T) + \text{nullity}(T) = n \Rightarrow n = \text{rank}(T) \leq \dim(W) = m.$$

□

41. Let T be a linear transformation on a vector space V and let $\dim V = n$. If

$$T^{n-1}(\mathbf{x}) \neq \mathbf{0} \text{ but } T^n(\mathbf{x}) = \mathbf{0} \text{ for some } \mathbf{x} \in V,$$

then show that the set $\{\mathbf{x}, T(\mathbf{x}), \dots, T^{n-1}(\mathbf{x})\}$ is a basis for V . Also, find the matrix representation of T with respect to this basis.

Solution: Let $a_1\mathbf{x} + a_2T(\mathbf{x}) + \dots + a_nT^{n-1}(\mathbf{x}) = \mathbf{0}$. Applying T^k on both sides of this equation, successively for $k = n-1, \dots, 2, 1$, we find that $a_1 = a_2 = \dots = a_{n-1} = a_n = 0$. Hence the vectors $\mathbf{x}, T(\mathbf{x}), \dots, T^{n-1}(\mathbf{x})$ are linearly independent.

We have

$$\begin{aligned} T(\mathbf{x}) &= 0\mathbf{x} + T(\mathbf{x}) + 0T^2(\mathbf{x}) + \dots + 0T^{n-1}(\mathbf{x}) \\ T(T(\mathbf{x})) &= 0\mathbf{x} + 0T(\mathbf{x}) + T^2(\mathbf{x}) + \dots + 0T^{n-1}(\mathbf{x}) \\ &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ T(T^{n-2}(\mathbf{x})) &= 0\mathbf{x} + 0T(\mathbf{x}) + 0T^2(\mathbf{x}) + \dots + T^{n-1}(\mathbf{x}) \\ T(T^{n-1}(\mathbf{x})) &= 0\mathbf{x} + 0T(\mathbf{x}) + 0T^2(\mathbf{x}) + \dots + 0T^{n-1}(\mathbf{x}). \end{aligned}$$

Hence the required matrix is $[a_{ij}]$, where

$$a_{ij} = \begin{cases} 1 & \text{if } i = j + 1, \\ 0 & \text{otherwise.} \end{cases}$$

□

42. Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, defined by

$$T[x, y, z]^t = [2x + 3y - z, 4x - y + 2z]^t \text{ for all } [x, y, z]^t \in \mathbb{R}^3.$$

Find $[T]_{C \leftarrow B}$, where $B = \{[1, 1, 0]^t, [1, 2, 3]^t, [1, 3, 5]^t\}$ and $C = \{[1, 2]^t, [2, 3]^t\}$.

Solution: We have

$$\begin{aligned} T([1, 1, 0]^t) &= [5, 3]^t = -9[1, 2]^t + 7[2, 3]^t \\ T([1, 2, 3]^t) &= [5, 8]^t = [1, 2]^t + 2[2, 3]^t \\ T([1, 3, 5]^t) &= [6, 11]^t = 4[1, 2]^t + [2, 3]^t. \end{aligned}$$

Hence

$$[T]_{C \leftarrow B} = [[T([1, 1, 0]^t))]_C, [T([1, 2, 3]^t))]_C, [T([1, 3, 5]^t))]_C = \begin{bmatrix} -9 & 1 & 4 \\ 7 & 2 & 1 \end{bmatrix}.$$

□

43. Let A be an $n \times n$ matrix. Find the eigenvalues of $A - 3I$ in terms of the eigenvalues of A . Also, show that their corresponding eigenspaces are equal.

Solution: From $|A - \lambda I| = |(A - 3I) - (\lambda - 3)I|$, we find that λ is an eigenvalue of A iff $\lambda - 3$ is an eigenvalue of $A - 3I$.

Let $\mathbf{x}$ be an eigenvector of A with corresponding eigenvalue λ . Then we have

$$(A - 3I)\mathbf{x} = A\mathbf{x} - 3\mathbf{x} = (\lambda - 3)\mathbf{x}.$$

Again if $\mathbf{y}$ is an eigenvector of $A - 3I$ with corresponding eigenvalue $\lambda - 3$, then we have

$$A\mathbf{y} = (A - 3I)\mathbf{y} + 3\mathbf{y} = (\lambda - 3)\mathbf{y} + 3\mathbf{y} = \lambda\mathbf{y}.$$

From the above, we have seen that $E_\lambda(A) = E_{\lambda-3}(A - 3I)$. □

44. Let $A = [a_{ij}]$ be an $n \times n$ matrix and let $k \in \mathbb{R}$. Suppose that $\sum_{j=1}^n a_{ij} = k$ for $i = 1, 2, \dots, n$. Show that k is an eigenvalue of A . Find an eigenvector of A corresponding to the eigenvalue k .

Solution: Let $\mathbf{u} = [1, 1, 1, \dots, 1]^t \in \mathbb{R}^n$. Then it is easy to see that $A\mathbf{u} = k\mathbf{u}$. Hence k is an eigenvalue of A and $\mathbf{u}$ is an eigenvector corresponding to k . □

45. Find all real values of a, b, c, d, e, f for which the matrix $\begin{bmatrix} 1 & a & b & c \\ 0 & 1 & d & e \\ 0 & 0 & 1 & f \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is diagonalizable.

Solution: Let the given matrix be A . Since all the eigenvalues of A are 1, if A is diagonalizable then there exists an invertible matrix T such that $T^{-1}AT = I$, which gives that $A = I$. Hence we must have $a = b = c = d = e = f = 0$ for the diagonalizability of A . □

46. Let A be a 6×6 matrix with characteristic polynomial $p(\lambda) = (1 + \lambda)(1 - \lambda)^2(2 - \lambda)^3$.
- (a) Prove that it is not possible to find three linearly independent vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^6$ such that $A\mathbf{v}_1 = \mathbf{v}_1, A\mathbf{v}_2 = \mathbf{v}_2$ and $A\mathbf{v}_3 = \mathbf{v}_3$.
- (b) If A is diagonalizable, find the dimensions of the eigenspaces $E_{-1}(A), E_1(A)$ and $E_2(A)$?

Solution:

- (a) Since the characteristic polynomial of A is $p(\lambda) = (1 + \lambda)(1 - \lambda)^2(2 - \lambda)^3$, 1 is an eigenvalue of A and the algebraic multiplicity of 1 is 2. Thus the geometric multiplicity of 1 (dimension of $E_1(A)$) can be at most 2. Therefore there cannot be three linearly independent vectors in $E_1(A)$. Hence there do not exist three linearly independent vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^6$ such that $A\mathbf{v}_1 = \mathbf{v}_1, A\mathbf{v}_2 = \mathbf{v}_2$ and $A\mathbf{v}_3 = \mathbf{v}_3$.
- (b) If A is diagonalizable, then the geometric multiplicities of each eigenvalue of A will be equal to the corresponding algebraic multiplicities. Hence $\dim E_{-1}(A) = 1, \dim E_1(A) = 2$ and $\dim E_2(A) = 3$. □

47. Let A and B be two $n \times n$ matrices satisfying $AB = BA$ and let B have n distinct eigenvalues. Show that the matrix A is diagonalizable.

Solution: Since B is an $n \times n$ matrix with n distinct eigenvalues, B must be diagonalizable. Let Q be an invertible matrix such that $QBQ^{-1} = D$, where D is a diagonal matrix. Then using $AB = BA$, we have

$$(QAQ^{-1})(QBQ^{-1}) = (QBQ^{-1})(QAQ^{-1}) \Rightarrow YD = DY, \text{ where } Y = QAQ^{-1}.$$

Let $Y = [y_{ij}]$ and let the diagonal entries of D be $d_1, d_2, \dots, d_n$. Since the diagonal entries of D are the n distinct eigenvalues of B , we find that $d_i \neq d_j$ for $i \neq j$. Now for $i \neq j$, we have

$$DY - YD = \mathbf{O} \Rightarrow (DY - YD)_{ij} = 0 \Rightarrow (d_i - d_j)y_{ij} = 0, 1 \leq i, j \leq n \Rightarrow y_{ij} = 0, \text{ since } d_i \neq d_j.$$

Thus Y is a diagonal matrix. Hence $Y = QAQ^{-1}$ implies that the matrix A is also diagonalizable. □
